

ORIGINAL ARTICLE

Combined non-pharmacological interventions for newborn pain relief in two degrees of pain procedures: A randomized clinical trial

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Conflicts of interest

None declared.

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Abstract

Background: Non-pharmacological interventions are effective neonatal pain reduction strategies. We aimed to study the effects of non-nutritive sucking (NNS) and swaddling on infants' behavioural and physiological parameters during shallow or deep heel stick procedures.

Method: In this prospective, multi-centred, randomized controlled clinical trial, we enrolled 671 newborns. The infants undergoing shallow or deep heel stick procedures were randomized into four groups: oral sucrose (routine care, group S), oral sucrose combined with NNS (group NS), oral sucrose combined with swaddling (group SS) and oral sucrose combined with NNS and swaddling (group NSS). The behavioural responses were evaluated by the Revised Neonatal Facial Coding System and the physiological signals were monitored by electrocardiogram monitors.

Results: A significant synergistic analgesic effect was observed between the NS and SS groups in both the shallow ($F = 5.952$, $p = 0.015$) and deep heel stick ($F = 7.452$, $p = 0.007$) procedure. NSS group exhibited the lowest pain score. For the deep heel stick procedure, the NS group had a significantly lower increase in heart rate (HR)% and decrease in $\text{SPO}_2\%$ than the S group ($F = 17.540$, $p = 0.000$, $F = 10.472$, $p = 0.001$), while this difference was not observed in the shallow heel stick procedure. No difference was found between the S and SS groups, in terms of different physiological parameters.

Conclusion: Non-nutritive sucking and swaddling had synergistic effects on pain relief when used with oral sucrose. For the deep heel stick procedure, oral sucrose combined with NNS and swaddling provided the best pain relief effect. For the shallow heel stick procedure, addition of NNS and swaddling did not improve the effects.

1. Introduction

Neonatal pain management is an identified need and appropriate measures are evolving. Non-pharmacologic interventions (NPIs), such as oral sucrose, non-nutritive sucking, swaddling, facilitated tucking and

kangaroo care are recommended for neonatal pain management (Gitto et al., 2012; Shah et al., 2012; McNair et al., 2013; Pillai Riddell et al., 2013; Johnston et al., 2014; Carbajal et al., 2015; Harrison et al., 2015). Oral sucrose has been suggested as the standard of care (Harrison et al., 2012; Stevens et al., 2013; Shen and El-Chaar, 2015). However, Curry

What's already known about this topic?

- Non-pharmacological interventions such as oral sucrose, non-nutritive sucking (NNS) and swaddling are effective neonatal pain reduction strategies. Few studies have measured the interaction between different non-pharmacological interventions or compared the interventions alone and in combination with procedures exhibiting different levels of pain.

What does this study add?

- This study demonstrated that oral sucrose combined with NNS as well as oral sucrose combined with swaddling had synergistic effects on reducing the neonatal pain score. For deep heel stick procedures, oral sucrose, NNS and swaddling combination showed the best effects; however, for shallow heel stick procedures, oral sucrose alone was enough to achieve an optimal analgesic effect.

and colleagues pointed out that oral sucrose does not show an effect on pain relief during the process of immunization (Curry et al., 2012). Grabska et al. (2004) also pointed out that sucrose did not provide pain relief when used for retinopathy screening. Furthermore, Milazzo and colleagues reported that no differences in heart rate (HR), oxygen saturation (O_2 Sat) and pain subscales was observed when sucrose was used to ease the pain during arterial punctures (Milazzo et al., 2011). Hence, oral sucrose alone may not be enough for these procedures, which are more painful than heel lance procedures.

Swaddling is known to shorten the time to return heart rate and oxygen saturation to baseline and decrease crying time (Ohgi et al., 2004). Non-nutritive sucking (NNS) has also been shown to reduce the behavioural pain response during circumcision (Bellieni et al., 2013), immunizations (Liaw et al., 2011) and heel lance procedures in full-term neonates (Liaw et al., 2010). Some studies have even demonstrated that combined NPIs provide better pain relief than single NPI (O'Sullivan et al., 2010; Cignacco et al., 2012; Naughton, 2013). A few other studies also point out that there are no differences between combined NPIs and single NPI (Carbajal et al., 2002; Harrington et al., 2012). However, the interaction effects of using different NPIs are not well studied.

Pain in the neonate can be classified into acute pain, established pain, and prolonged pain (Anand

et al., 2005). Acute pain occurs from skin-breaking procedures or tissue injury caused by diagnostic or therapeutic interventions. Different procedures cause pain to different degrees. For example, heel stick procedures were proven to cause much more pain than venipuncture (Shah and Ohlsson, 2011). Also, lancing pain procedures were mainly related to lancet penetration depth, with the smaller lancet penetration depth causing less injury to the tissue and therefore less pain (Fruhstorfer et al., 1995; Kocher et al., 2009). The lancet diameter is thought to be of minor importance for lancing pain; however, when the penetration depth reached 1.2 mm and lancets were thicker, they resulted in more pain (Fruhstorfer et al., 1999). Since pain varies for different procedures, they should be treated accordingly. Administering combined NPIs require more personnel and time than a single NPI; however, a better understanding of which procedure would require single versus combined NPIs would be critical for neonatal care.

We studied the pain relief effect of NNS and swaddling, and their interaction effect when oral sucrose was taken as a routine pain intervention in shallow and deep heel stick procedures. We hypothesized that a single NPI would be sufficient for pain reduction in lighter pain procedures (referring to shallow heel stick procedures), while a combination of NPIs would provide better pain relief when it came to heavier pain procedures (referring to deep heel stick procedures).

2. Methods

This study employed oral sucrose as routine care to study the effectiveness of NNS, swaddling, and the interaction effect between NNS and swaddling on pain relief in full-term neonates undergoing heel-stick procedures for blood glucose testing (shallow heel stick procedure) and neonatal congenital metabolism disease screening (deep heel stick procedure). Patients were divided into four groups: oral sucrose (group S), NNS with oral sucrose (group NS), swaddling with sucrose (group SS), and the combination of NNS and swaddling with oral sucrose (group NSS). The pain response was measured by the revised Neonatal Facial Coding System (the revised NFCS) score and physiological signals, such as oxygen saturation levels (O_2 Sat) and HR changes monitored by electrocardiogram monitors.

2.1 Randomized multicentre controlled trial

This was a prospective, randomized, multicentre, controlled trial designed to study the analgesic effect

of NNS and swaddling, and their effects when combined with oral sucrose in full-term infants undergoing shallow or deep heel stick procedures. The Institutional Review Board of Children's Hospital Chongqing Medical University approved the study.

2.2 Study sample and group randomization

Based on the frequency of the heel stick applications and purpose of the heel stick in each hospital, we enrolled patients from the Newborn Intensive Care Unit, Full Term Neonates Department, Children's Hospital of Chongqing Medical University, Chongqing, and Hunan Children's Hospital, Hunan for the deep heel stick procedure. Patients from the Newborn Intensive Care Unit, Full Term Neonates Department, Shenzhen Children's Hospital, Shenzhen, and Chengdu Women's & Children's Central Hospital, Chengdu were enrolled in the shallow heel stick procedure. Inclusion criteria were: (1) gestational age between 37 and 42 weeks at delivery; (2) postnatal age between 3 and 28 days; (3) birth-weight between 2500 g and 4000 g; (4) Apgar score ≥ 8 at 5 min after birth; (5) resting HR between 120 and 140 per/min and resting O_2 Sat $\geq 95\%$; and (6) ordered for neonatal congenital metabolism disease screening or blood glucose test. Exclusion criteria were: (1) history of surgery; (2) congenital or known genetic anomalies; (3) disorders affecting brain circulation or cardiovascular system; (4) documented congenital or nosocomial sepsis; (5) severe medical conditions requiring treatment such as sedatives, muscle relaxants, anti-epileptic, or analgesic drugs; (6) in the condition of fasting; and (7) mother with a history of substance-abuse.

2.3 Study procedures

The physician assistant informed the infant's parent or legal guardian of the purpose of the study. If the parent or guardian consented to participation in the study, an informed written consent was obtained. Study infants were randomly assigned to one of four study groups by simple randomization. Infants were numbered in sequence as 001, 002, 003.....800 according to the sequence they were accepted into the study. If the infant qualified and his/her guardian agreed to participate, his/her ID number was sent to the randomization schedule implementer (a statistical expert from Chongqing Medical University) to make groups. Each infant was assigned a random digit by using a random number table generated with SPSS19.0. The randomization codes were kept in a secure location that could not be accessed by

study personnel. A simple calculation utilizing the random digit was used to determine a remainder (remainder = random digit/4), which was then used to decide which group the infant belonged to. For example, if the remainder was 1, then the infant was assigned to group S. If the remainder was 2, then the infant was assigned to group NS. If the remainder was 3 then the infant was assigned to group SS. If the remainder was 0 then the group was assigned to NSS. The interventions in the four groups were as follows:

Group S: 2 ml of 24% sucrose was administered to the infant's mouth by syringe 2 min before the heel-stick procedure.

Group NS: 2 ml of 24% sucrose was administered to the infant's mouth by syringe 2 min before the heel stick procedure, and then a standard silicone newborn pacifier was placed into the infant's mouth until the end of the process.

Group SS: infants were swaddled with a cotton blanket, upper but not lower limb movements were restricted by the blanket, and then 2 ml of 24% sucrose was administered to the infant's mouth by syringe 2 min before the heel stick procedure. The lower limbs were swaddled right after the heel stick procedure until the end of the process.

Group NSS: infants were swaddled with a cotton blanket, upper but not lower limb movements were restricted by the blanket, then 2 ml of 24% sucrose was administered into the infant's mouth by syringe before the heel stick procedure, then a standard silicone newborn pacifier was placed into the infant's mouth, the lower limbs were swaddled right after the heel stick procedure until the end of the process.

Procedures were performed 1 h after feeding. Neonates from S and NS groups, who were wearing baby clothing, were placed under the radiant warmers before heel stick procedure with the temperature set to 36 °C. Neonates from the SS and NSS groups who were swaddled with the cotton blanket were not exposed to warm temperatures during this procedure.

Four nurses from four centres were trained to perform the heel stick procedures, which included how to specifically locate the site of the heel stick, how to squeeze the foot to collect blood and swaddle the infant. Their evaluation criteria included obtaining one trial success rate for heel stick with the ability to swaddle the infant's lower limbs within 5 s. Another four nurses from four centres were trained to perform non-pharmacological strategies, which included how to specifically administer sucrose into the infants' mouth and how to swaddle the infant. Their

evaluation criteria included administering 2 ml of sucrose into the infant's mouth within 30 s without the infant buckling and swaddling the infant using standard procedures within 10 s. These eight nurses were blinded to the study and the infants' clinical information.

2.4 Outcome measures

Outcome variables included the infants' behavioural and physiological response (HR and O₂ Sat) to the procedure. The pain response was assessed by a research assistant trained to use the revised NFCS. The revised NFCS included five criteria such as (1) brow bulge; (2) eye squeeze; (3) naso-labial furrow; (4) horizontal mouth; and (5) taut tongue, which code only for the presence or absence of these facial actions. Each facial action is recorded as 'occurring' (score 1) or 'not occurring' (score 0), with a maximum score of 5 and minimum score of 0. A higher score related to more pain (Supporting Information Table S1; Grunau and Craig, 1987; Peters et al., 2003; Arias and Guinsburg, 2012). The research assistant was trained by the corresponding author using videotapes of heel stick procedures on full-term infants. Twenty videotapes were randomly chosen to determine coding reliability between the research assistant and the corresponding author, whose inter-rater reliability was 0.89–0.94 and intra-rater reliability was 0.95–0.98. The research assistant then coded all 671 videotapes. During the coding process, the corresponding author assessed the inter-rater reliability with the research assistant during every 67th videotape and kept the reliability $\geq 85\%$. The research assistant also independently re-evaluated her intra-rater reliability every 30th videotape and maintained the intra-rater reliability $\geq 95\%$.

HR and O₂ Sat levels were recorded from the videotape. Each videotape was named after the infant's hospital ID number when it was sent back to research team, and kept in the same sequence of 001...671 (according to randomization). ECG tracings were monitored regularly and calibrated by technicians. Video imaging equipment was checked prior to data collection. No adverse events were observed during the procedure.

Time-triggered coding was used to measure the pain score and readings of HR and O₂ Sat levels. The research assistant coded the pain score, HR and O₂ Sat levels at 10 s intervals at baseline (last 30 s of the 2 min before heel stick) and during the procedural phases (the last 30 s of the 1 min after heel-stick). The average pain score, HR and O₂ Sat levels

for the two phases were calculated. Since infants were quiet during the baseline phase, the pain scores were 0, thus average pain scores were only calculated during the procedural phase. The average increase in HR percentage ($\uparrow\text{HR}\%$) and average decrease in O₂ Sat percentage ($\downarrow\text{SPO}_2\%$) was calculated for statistical analyses.

Videotapes were stored on a mobile hard disk drive and mailed to the research team at Children's Hospital of Chongqing Medical University. The subject ID number, gender, gestational age, postnatal age, birthweight, Apgar score, resting heart rate, oxygen saturation levels, and any specific illness was recorded manually by the physician assistants and mailed to the research teams after all of the data from the videotapes was determined to reveal intervention groups and centres.

2.5 Statistical analysis

2.5.1 Power analysis

The power of the study was estimated using GPOWER version 3.1.2, and calculated based on ANOVA: fixed effects, special, main effects and interactions among the four treatment conditions. For average pain scores, the Sum of Squares (SS) for the corrected model and total were 10.94 and 170.22, respectively, based on the effect of size, which was 0.26 ($\eta^2 = \frac{SS_{\text{within groups}}}{SS_{\text{total}}}$). The effect size (0.26), total sample size (293), groups (4) and a significance level of 0.05 led to a study power of 0.96. For $\Delta\text{SPO}_2\%$, the Sum of Squares for the corrected model and total were 34.28 and 886.71, respectively, based on the effect size, which was 0.19. The effect size (0.19), total sample size (293), groups (4) and a significance level of 0.05 led to a study power of 0.89. For $\Delta\text{HR}\%$, the Sum of Squares for the corrected model and total were 1198.22 and 18889.36, respectively, based on the effect size, which was 0.25. The effect size (0.25), total sample size (293), groups (4) and a significance level of 0.05 led to a study power of 0.95.

2.6 Statistical methodology

Demographic variables such as gender were evaluated for significant differences between the four intervention groups using the χ^2 analysis. For the remaining demographic variables such as gestational age, postnatal age and Apgar scores, Kruskal–Wallis analysis was performed. For weight at birth and time for blood collection, a one-way ANOVA test was used to determine if there were significant differ-

ences among the intervention groups. Factorial analysis was performed to evaluate interaction and main effect among the intervention groups, procedures and hospitals. Multi-variable linear regression analysis was performed to determine the influential factors on $\uparrow$ HR%, $\downarrow$ SPO₂% and pain score. For all of the statistical tests, a p value of <0.05 was considered significant.

3.. Results

Patients were recruited from June 25, 2012 through February 25, 2013. As indicated in Fig. 1, 1800 full-term newborns were screened for participation in the study. About 70.88% (1276) met study criteria, and of those 96.31% (1229) agreed to participate and 54.60% (671) of them received the heel stick procedure to collect blood. Of the 671 infants, 293 received the shallow heel stick procedure and 378 received the deep heel lance procedure. For the shallow and deep heel stick procedures, 84 and 92 were randomly assigned to group S, respectively, 76 and 91 to group NS, respectively, 72 and 95 to group SS, respectively, as well as 61 and 100 to group NSS, respectively. Demographic variables did not vary significantly among the study groups (Supporting Information Table S2). Supporting Information Table S2.1 showed the reason for infant hospitalization, while

Supporting Information Table S2.2 demonstrated general statistical data for the infants in each hospital.

Factorial analyses on pain scores showed that the SS and NS groups had an obvious interaction effect ($p < 0.05$). Interestingly, no significant differences were found among the different hospitals, and no obvious interaction effects were found between procedures and NNS or swaddling (Table 1).

Factorial analysis on $\uparrow$ HR% and $\downarrow$ SPO₂% showed that significant differences were found among the different hospitals and obvious interaction effects were found between the effects of different procedures and NNS versus swaddling ($p < 0.05$) (Tables 2 and 3). Therefore, an advanced variance of analysis was performed for shallow and deep heel stick procedures separately, which showed that $\uparrow$ HR% was significantly lower in the NS versus S group for deep heel stick procedures ($p < 0.05$) (Table 2.1), but not for shallow heel stick procedures (Table 2.2). Table 3.1 showed that $\downarrow$ SPO₂% was significantly lower in the NS versus S group for deep heel stick procedures ($p < 0.05$), but not in shallow heel stick procedures (Table 3.2). No significant differences were found between S and SS groups.

Multi-variable linear regression analysis showed that the procedures, gender, postnatal age, birth-weight, gestational age and time for blood collection

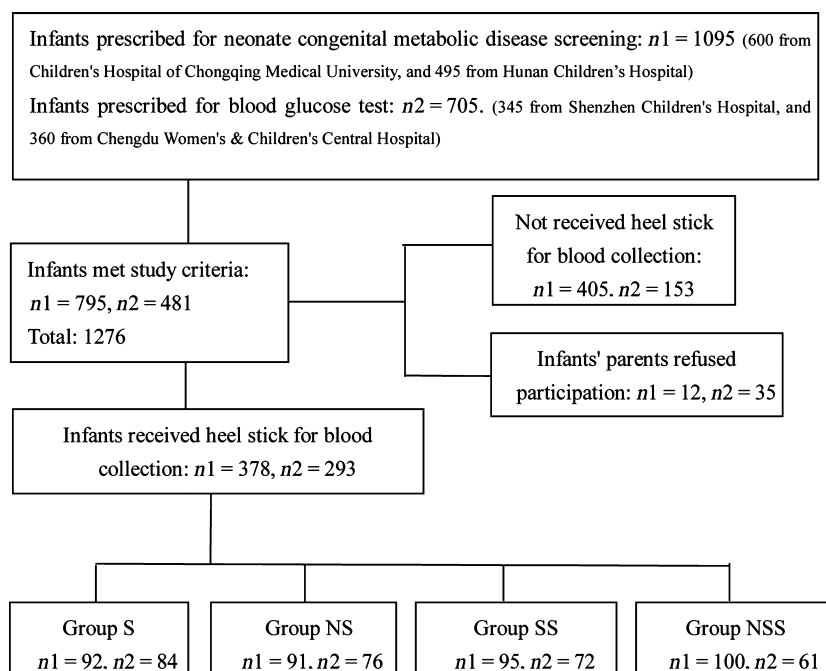


Figure 1 Randomization table.

Table 1 Factorial analyses on pain score.

	Pain score	<i>F, p</i>
X1	0.11 ± 0.68	
X2	0.17 ± 0.81	X1: <i>F</i> = 0.16, <i>p</i> = 0.9261
X3	0.16 ± 0.75	
X4	0.13 ± 0.74	q2: <i>F</i> = 12.12, <i>p</i> = 0.0005
Group S	0.45 ± 1.34	q3: <i>F</i> = 16.21, <i>p</i> < 0.0001
Group NS	0.02 ± 0.17	X1*q3: <i>F</i> = 0.09, <i>p</i> = 0.9675
Group SS	0.05 ± 0.36	X1*q2: <i>F</i> = 0.15, <i>p</i> = 0.9287
Group NSS	0.02 ± 0.20	q2*q3: <i>F</i> = 13.31, <i>p</i> = 0.0003

Note: Data were presented as Mean ± SD. X1: compare between hospitals, q2: comparison between swaddling with and without (group SS and group S), q3: comparison between NNS with and without (group NS and group S), X1*q2: interaction effect analysis between hospitals and swaddling, X1*q3: interaction effect analysis between hospitals and NNS, q2*q3: interaction effect analysis between swaddling and NNS.

Table 2 Factorial analyses on increased heart rate (HR) %.

	↑HR%	<i>F, p</i>
X1	16.89 ± 9.85	
X2	15.56 ± 7.99	X1: <i>F</i> = 18.38, <i>p</i> < 0.0001
X3	11.42 ± 8.43	
X4	14.00 ± 8.90	q2: <i>F</i> = 1.32, <i>p</i> = 0.2508
Group S	15.49 ± 9.09	q3: <i>F</i> = 14.65, <i>p</i> = 0.0001
Group NS	13.20 ± 8.39	X1*q3: <i>F</i> = 3.52, <i>p</i> = 0.0148
Group SS	14.92 ± 9.73	X1*q2: <i>F</i> = 7.75, <i>p</i> < .0001
Group NSS	12.24 ± 7.96	q2*q3: <i>F</i> = 0.11, <i>p</i> = 0.7450

Note: Data were presented as Mean ± SD. X1: comparison between hospitals, q2: comparison between swaddling with and without (group SS and group S), q3: comparison between NNS with and without (group NS and group S), X1*q2: interaction effect analysis between hospitals and swaddling, X1*q3: interaction effect analysis between hospitals and NNS, q2*q3: interaction effect analysis between swaddling and NNS.

Table 2.1 Factorial analysis on increased heart rate (HR) % in deep heel stick procedure.

	Swaddling		<i>F, p</i>
	–	+	
–	Group S (<i>n</i> = 92)	Group SS (<i>n</i> = 95)	
NNS	15.51 ± 9.08	16.33 ± 11.04	q3: <i>F</i> = 17.540, <i>p</i> = 0.000
+	Group NS (<i>n</i> = 91)	Group NSS (<i>n</i> = 100)	
	12.44 ± 9.06	12.02 ± 8.15	q2: <i>F</i> = 0.024, <i>p</i> = 0.877
			q2*q3: <i>F</i> = 0.405, <i>p</i> = 0.52

were not influential factors on the pain score, ↑HR% and ↓SPO₂%. Swaddling and NNS were the influential factors on pain score and ↑HR% (Tables 4 and 5). Table 6 showed that swaddling, NNS and

Table 2.2 Factorial analysis on increased heart rate (HR) % in shallow heel stick procedure.

	Swaddling		<i>F, p</i>
	–	+	
–	Group S (<i>n</i> = 84)	Group SS (<i>n</i> = 72)	
NNS	14.27 ± 8.18	13.12 ± 7.29	q3: <i>F</i> = 1.069, <i>p</i> = 0.302
+	Group NS (<i>n</i> = 76)	Group NSS (<i>n</i> = 61)	
	14.11 ± 7.46	12.61 ± 7.70	q2: <i>F</i> = 0.991, <i>p</i> = 0.320
			q2*q3: <i>F</i> = 0.162, <i>p</i> = 0.687

Note: Data were presented as Mean ± SD. q2: comparison between swaddling with and without (group SS and group S), q3: comparison between NNS with and without (group NS and group S), q2*q3: interaction effect analysis between swaddling and NNS.

Table 3 Factorial analysis on decreased SPO₂%.

	↓SPO ₂ %	<i>F, p</i>
X1	2.20 ± 1.91	
X2	1.80 ± 1.88	X1: <i>F</i> = 14.95, <i>p</i> < 0.0001
X3	1.09 ± 1.52	
X4	1.39 ± 1.55	q2: <i>F</i> = 1.01, <i>p</i> = 0.2055
Group S	2.00 ± 1.63	q3: <i>F</i> = 14.01, <i>p</i> = 0.0002
Group NS	1.52 ± 1.91	X1*q3: <i>F</i> = 2.95, <i>p</i> = 0.0323
Group SS	1.70 ± 1.86	X1*q2: <i>F</i> = 3.22, <i>p</i> = 0.0223
Group NSS	1.21 ± 1.60	q2*q3: <i>F</i> = 0.07, <i>p</i> = 0.7927

Note: Data were presented as Mean ± SD. X1: comparison between hospitals, q2: comparison between swaddling with and without (group SS and group S), q3: comparison between NNS with and without (group NS and group S), X1*q2: interaction effect analysis between hospitals and swaddling, X1*q3: interaction effect analysis between hospitals and NNS, q2*q3: interaction effect analysis between swaddling and NNS.

Table 3.1 Factorial analysis on decreased SPO₂% in deep heel stick procedure.

	Swaddling		<i>F, p</i>
	–	+	
–	Group S (<i>n</i> = 92)	Group SS (<i>n</i> = 95)	
NNS	1.99 ± 1.63	1.81 ± 1.93	q3: <i>F</i> = 10.472, <i>p</i> = 0.001
+	Group NS (<i>n</i> = 91)	Group NSS (<i>n</i> = 100)	
	1.51 ± 1.90	1.22 ± 1.67	q2: <i>F</i> = 1.970, <i>p</i> = 0.161
			q2*q3: <i>F</i> = 0.069, <i>p</i> = 0.793

hospitals were the influential factors on ↓SPO₂%. Children's Hospital of Chongqing Medical University and Hunan Children's Hospital had higher ↓SPO₂% compared with Shenzhen Children's Hospital and Chengdu Women's & Children's Central Hospital.

Table 3.2 Factorial analysis on decreased SPO₂% in shallow heel stick procedure.

Swaddling		<i>F</i> , <i>p</i>
–	+	
–	Group S (<i>n</i> = 84)	Group SS (<i>n</i> = 72)
NNS	2.00 ± 1.65	1.57 ± 1.75
+	Group NS (<i>n</i> = 76)	Group NSS (<i>n</i> = 61)
	1.53 ± 1.93	1.19 ± 1.49
	q2: <i>F</i> = 2.266, <i>p</i> = 0.133	
	q3: <i>F</i> = 3.696, <i>p</i> = 0.056	
	q2*q3: <i>F</i> = 0.037, <i>p</i> = 0.849	

Note: Data were presented as Mean ± SD. q2: comparison between swaddling with and without (group SS and group S), q3: comparison between NNS with and without (group NS and group S), q2*q3: interaction effect analysis between swaddling and NNS.

Table 4 Multivariable linear regressions analysis on pain score.

Variable	Parameter estimate	Standard error	<i>t</i> value	<i>p</i> value	Standardized estimate
Constant	0.36002	0.04824	7.46	<0.0001	0
q2	–0.20173	0.05674	–3.56	0.0004	–0.13456
q3	–0.23661	0.05673	–4.17	<0.0001	–0.15783

4. Discussion

We used oral sucrose as a routine pain relief measure and demonstrated that a combination of NNS and swaddling had a synergistic effect on reducing the pain score. Swaddling effectively reduced pre-term infant distress with better motor organization and more self-regulatory ability (Shu et al., 2014; da Motta Gde and da Cunha, 2015). NNS moderating pain-related reactivity and immediate regulation of pain in neonates have been previously explained (Liaw et al., 2010, 2011; Meek and Huertas, 2012). When these two interventions were combined together with a standard pain reducing strategy of oral sucrose, the pain-related behaviour was strikingly reduced. This is the first study to explore the possible synergistic effects between NNS and swaddling along with oral sucrose as a routine intervention for neonatal pain.

Previous studies have focused on the interaction between oral sucrose and NNS. Boyle et al. reported findings from their pain procedure of retinopathy screening, where randomized infants underwent one of four interventions: sterile water with a pacifier, sucrose with a pacifier, sterile water or sucrose. In this case, they did not observe synergistic effects of NPI interventions on lowering pain scores (Boyle et al., 2006). Similar results were also found in a study by Blass and Watt, which measured crying duration, grimace and HR changes during a heel

Table 5 Multivariable linear regressions analysis on increased heart rate (HR) %.

Variable	Parameter estimate	Standard error	<i>t</i> value	<i>p</i> value	Standardized estimate
q2	0.09674	0.00808	11.97	<0.0001	0.40701
q3	0.07986	0.00807	9.90	<0.0001	0.33647

Table 6 Multivariable linear regressions analysis on decreased SPO₂%.

Variable	Parameter estimate	Standard error	<i>t</i> value	<i>p</i> value	Standardized estimate
a1	0.01782	0.00160	11.15	<0.0001	0.38587
a2	0.01452	0.00168	8.66	<0.0001	0.28628
q2	0.00532	0.00126	4.21	<0.0001	0.15443
q3	0.00281	0.00128	2.19	0.0288	0.08154

lance procedure (Blass and Watt, 1999). A study by Carbajal et al. also reported this when measuring pain during the subcutaneous injection of erythropoietin (Carbajal et al., 2002). However, a review on this issue indicated that NNS and oral sucrose may have an interaction effect (Naughton, 2013). Unfortunately, limited by our study design, we were not able to analyse the interaction effects between oral sucrose and NNS, but there is sufficient evidence to support that oral sucrose combined with NNS had a better effect on pain relief than their single use (Tsao et al., 2008; Elserafy et al., 2009; Dilli et al., 2014; Richard and Anand, 2014; Ucar et al., 2014). Our study supports this conclusion since we showed that during the deep heel stick procedure, NNS combined with oral sucrose lowered HR and O₂ saturation fluctuations when compared to oral sucrose alone.

In the shallow heel stick procedure, we did not find significant differences between HR and O₂ Sat levels between the NS and S groups. This finding supported evidence that oral sucrose alone would be enough to ease pain from the shallow heel stick procedure and that the addition of other NPIs does not further maximize this pain relief.

No significant difference in ↑HR% and ↓SPO₂% was observed between the S and SS groups. These findings suggested that sucrose with or without swaddling makes no obvious difference in sustaining physiological status, but combined with swaddling did reduce pain score and shorten times back to baseline heart rate and oxygen saturation levels (Ohgi et al., 2004). Thus, we believe that swaddling mainly sustains a stable behaviour.

In China, the practice of analgesia during procedures like heel lance, venipuncture and subcutaneous or intramuscular injection are rarely practised. Even in lumbar puncture procedures, the use of analgesics

may not be always employed. Recent studies in China showed that only 37.17% nurses gave correct answers to pain management knowledge, and most of the respondents thought NPIs could not ease or relieve pain (Qian et al., 2012; Wu, 2013). Factors that blocked implementation of pain management strategies included resistance to change, lack of knowledge and lack of time (Cong et al., 2014). This study was conducted in four neonatal departments of different tertiary hospitals located in major cities of China demonstrating the effect of pain relief using NPIs. The results from our study provide new insights and guidelines into sensitizing health care providers regarding newborn pain in China, and make it clear of when to use single NPIs versus combined NPIs during neonatal procedures. Though the administration of combined NPIs needs more personnel and time, they effectively reduce pain and optimum pain management should be the goal. Hence, appropriate guidelines need to be formulated for day-to-day clinical practice involving painful procedures in neonates.

The present study has several limitations. To measure the interaction effect between NNS and swaddling, the best design for interventions would be a blank control, NNS, swaddling and NNS combined with swaddling, but it is not ethical to withhold care from an infant in pain. Hence, we used oral sucrose as the standard of care, which has made it difficult to interpret whether an interaction exists between NNS and swaddling when oral sucrose is not involved. Even though we had designed strict inclusion and exclusion criteria for our study objectives, and tried to standardize every step of the process, slight differences may still exist when it comes to different centres. Although we made efforts to limit bias from the coder, there is a possibility that the coder could still have distinguished the different groups by assessing with or without NNS. Video capturing the neonatal face and ECG monitor in one image required a larger screen, large screen, which might have influenced coding subtle changes in facial expression. In addition, we had a disproportionate number of neonates in the group NSS when compared with groups S, NS or SS in the shallow heel stick procedure due to random error.

5. Conclusions

The present study demonstrated that NNS and swaddling had synergistic analgesic effects when combined with oral sucrose. For deep heel stick procedures, oral sucrose, NNS and swaddling were shown

to be the most effective when used in combination. For shallow heel stick procedures, oral sucrose alone was sufficient. Further studies will be required to determine whether these findings are reproducible.

Author contributions

X.Z. conceived the study, participated in its design, trained all the members on how to perform the procedures and helped to draft the manuscript. H.L. participated in the design of the study, coordination of all of the centres, monitored the quality of the procedures and drafted the manuscript. L.Y. performed statistical analysis. X. Z., H.H., G.T., Q.F. and S.S. carried out the procedures in the different centres.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's web-site:

Table S1. Revised Neonatal Facial Coding System.

Table S2. Baseline characteristics of participating subjects.

Table S2.1. Reasons for hospitalization.

Table S2.2. General statistical data in each hospital.

Note

Variable declaration part: X1 (X1: Children's Hospital of Chongqing Medical University, X2: Hunan Children's Hospital, X3: Shenzhen Children's Hospital, X4: Chengdu Women's & Children's Central Hospital.), dummy variable q1: procedures (shallow heel stick or deep heel stick), dummy variable: a1–a4 hospitals (a1: Children's Hospital of Chongqing Medical University, a2: Hunan Children's Hospital, a3: Shenzhen Children's Hospital, a4: Chengdu Women's & Children's Central Hospital.), q2: swaddling (with or without), q3: NNS (with or without), q4: gender (male or female), q5: postnatal age, q6: birthweight, q7: gestational age, q8: time for blood collection.